VANISHING ALONG AN AFFINE BASE CHANGE IN THE VANISHING SITUATION, WITH THE GLOBAL SECTIONS OF THE BASE CHANGED STRUCTURE SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Vanishing of the higher direct images of an affine morphism, and the Leray spectral sequence. Same chapter spaces-cohomology.tex, label lemma-affine-vanishing-higher-direct-images (invoked at proof line 3053) and Cohomology on Sites, label sites-cohomology-lemma-Leray (invoked at proof line 3054). They transport the vanishing of H^1 from the base to any space affine over it, which is part (1).
- Affineness is preserved by base change, and morphisms of affine schemes are affine. Morphisms of Spaces, label spaces-morphisms-lemma-base-change-affine (invoked at proof line 3057). Applied to the structure morphism $X \to \Spec(A)$ and the affine base change $\Spec(A') \to \Spec(A)$, it shows that $X' \to X$ is affine.
- The pushforward of the structure sheaf along an affine base change. Same chapter spaces-cohomology.tex, label lemma-affine-base-change (invoked at proof line 3062). It computes $(X' \to X)_*\mathcal{O}_{X'} = A' \otimes_A \mathcal{O}_X$, which turns the global sections computation into a tensor product over A.
- Global sections of the pullback of a module in the vanishing situation. Same chapter spaces-cohomology.tex, label lemma-vanishing-compute (invoked at proof line 3067). It gives $\Gamma(X, A' \otimes_A \mathcal{O}_X) = A'$, completing the second assertion.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07U8: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1902–1902. Navigation pointer for \ref{lemma-affine-base-change} used inside the counted span (line 3062); the argument that consumes it is retained in the span.

Source reference. Original lemma 073H: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1522–1522. Navigation pointer for \ref{lemma-affine-vanishing-higher-direct-images} used inside the counted span (line 3053); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UZ: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3009–3009. Navigation pointer for \ref{lemma-vanishing-compute} used inside the counted span (line 3068); the argument that consumes it is retained in the span.

Source reference. Original lemma 0732: sources/stacks-chow/tags.txt, source lines 9046–9046. Navigation pointer for \ref{sites-cohomology-lemma-Leray} used inside the counted span (line 3055); the argument that consumes it is retained in the span.

Source reference. Original reference 07UY: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2991–2991. Navigation pointer for \ref{situation-vanishing} used inside the counted span (line 3042); the argument that consumes it is retained in the span.

Source reference. Original lemma 03WI: sources/stacks-chow/tags.txt, source lines 5113–5113. Navigation pointer for \ref{spaces-morphisms-lemma-base-change-affine} used inside the counted span (line 3058); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 07UY.*

(1) *Given an affine morphism $X' \to X$ of algebraic spaces, we have $H^1(X', \mathcal{F}') = 0$ for every quasi-coherent $\mathcal{O}_{X'}$-module $\mathcal{F}'$.*
(2) *Given an A-algebra A' setting $X' = X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A')$ the morphism $X' \to X$ is affine and $\Gamma(X', \mathcal{O}_{X'}) = A'$.*

Proof. Part (1) follows from Lemma 073H and the Leray spectral sequence (Cohomology on Sites, Lemma 0732). Let $A \to A'$ be as in (2). Then $X' \to X$ is affine because affine morphisms are preserved under base change (Morphisms of Spaces, Lemma 03WI) and the fact that a morphism of affine schemes is affine. The equality $\Gamma(X', \mathcal{O}_{X'}) = A'$ follows as $(X' \to X)_*\mathcal{O}_{X'} = A' \otimes_A \mathcal{O}_X$ by Lemma 07U8 and thus

$$\Gamma(X', \mathcal{O}_{X'}) = \Gamma(X, (X' \to X)_*\mathcal{O}_{X'}) = \Gamma(X, A' \otimes_A \mathcal{O}_X) = A'$$

by Lemma 07UZ. □